

UNIVERSITI TUNKU ABDUL RAHMAN
Faculty of Information and Communication Technology



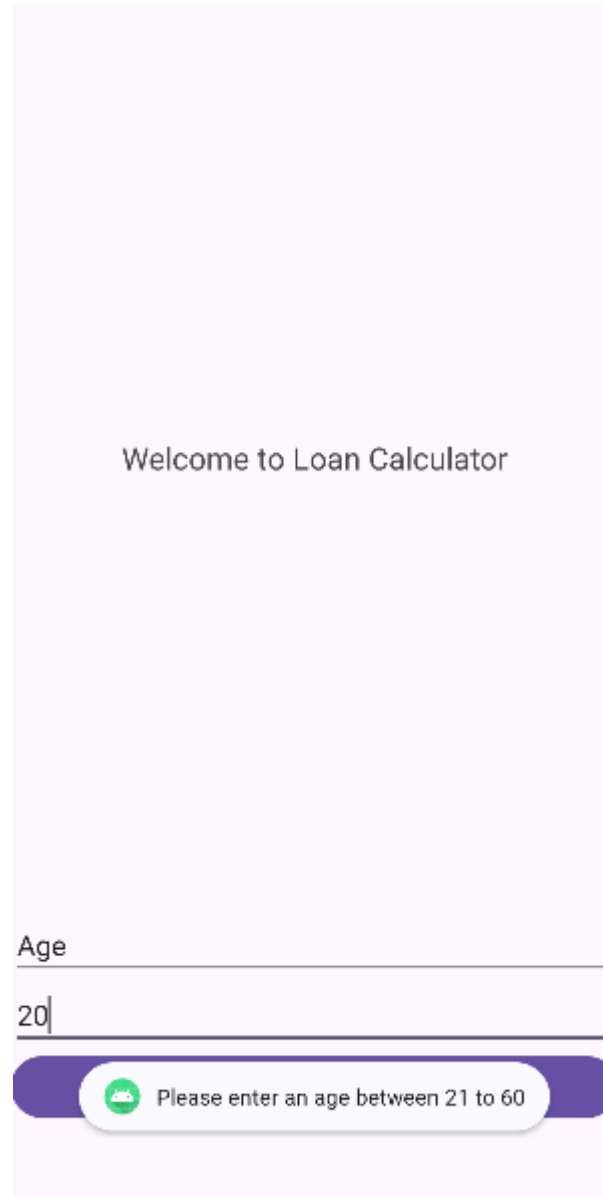
UCCD3223 Mobile Applications Development
(June 2024 Trimester)

Individual Practical Assignment

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Course	CN
Practical Group	P2
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Marking scheme	Marks	Remarks
Correctness of output	× 2.5	
Design of UI	× 2.5	
User Friendliness	× 3	
Neat Program Documentation		
Report Format		
TOTAL		

Homepage of Loan Calculate



Welcome to Loan Calculator

Age

20

Please enter an age between 21 to 60

Figure 1

According to the bank in Malaysia, the minimum age for having personal loan or any loan is 21 years old and above and maximum of the age will be 60 years old. Therefore, when the user enter age below 21 and above 60. It will pop out an alert message (**Figure 1**). After popped out the message, user can enter the correct age again. Then it will pop out an alert message “Submitted successfully” shown as **Figure x**.

Selection Page of Loan Calculate

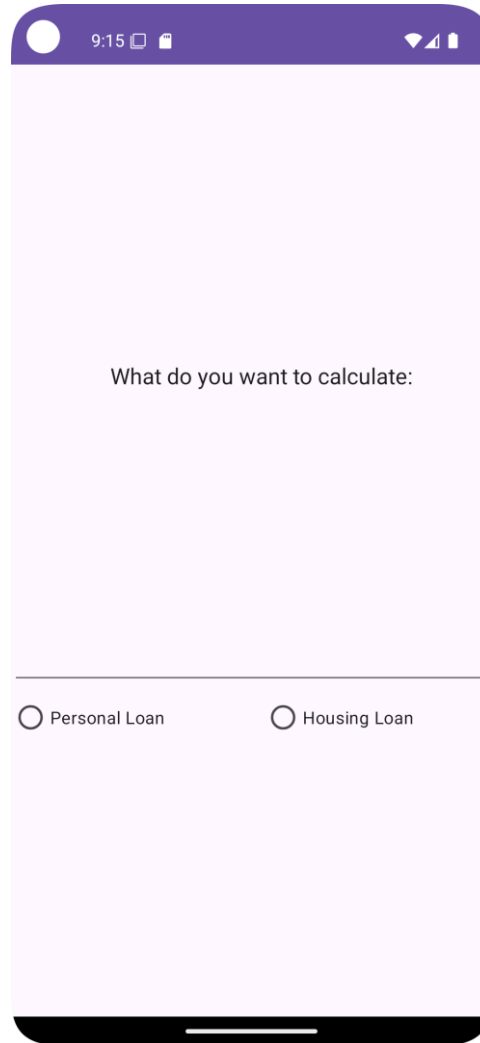
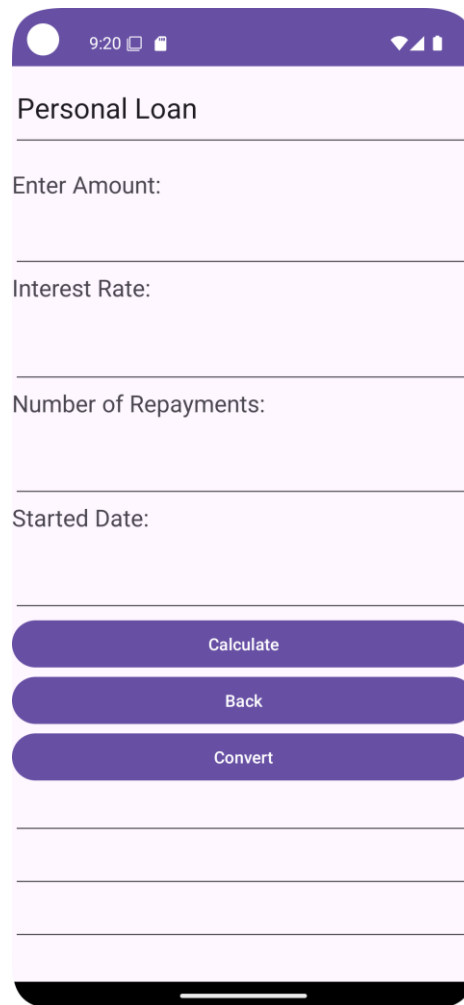
A mobile application interface for loan calculation. The screen has a light purple background. At the top, there is a dark purple status bar with a white circle icon, the time '9:15', and battery and signal icons. Below the status bar, the text 'What do you want to calculate:' is centered. At the bottom, there are two radio button options: 'Personal Loan' and 'Housing Loan'. The 'Personal Loan' option is selected, indicated by a small white dot inside the radio button circle. The 'Housing Loan' option is unselected, indicated by an empty radio button circle. The bottom of the screen shows a black home indicator bar.

Figure 2

After user enter their age on the Figure 1, it will proceed to the next page will is the selection as shown as **Figure 2**. In this section, user may choose one of the following which is Personal Loan and Housing Loan.

Personal Loan Calculation page



The screenshot shows a mobile application interface for a 'Personal Loan' calculator. The app has a purple header bar with a status bar at the top showing the time as 9:20 and various icons. The main title 'Personal Loan' is centered at the top of the form. Below the title, there are four input fields, each with a label and a light purple border: 'Enter Amount:', 'Interest Rate:', 'Number of Repayments:', and 'Started Date:'. At the bottom of the form, there are three stacked, rounded rectangular buttons with a dark purple background and white text: 'Calculate', 'Back', and 'Convert'. Below these buttons are three more empty, light purple rectangular boxes. The entire form is set against a light pink background.

Figure 3

When the user clicked on the Personal Loan button in **Figure 2**, it will proceed to personal loan calculation page. Then the user can enter the loan amount, interest rate, number of repayments and the started date which provides in **Figure 3**.

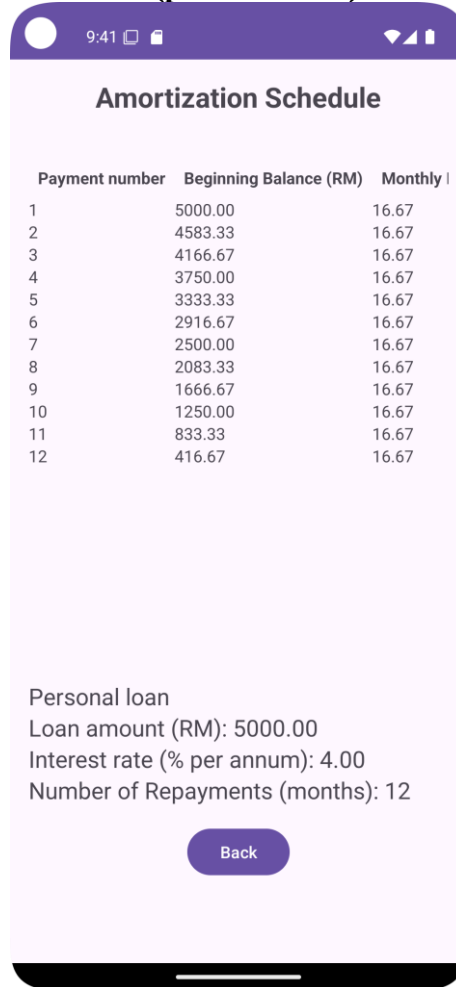
The screenshot shows a mobile application interface for a 'Personal Loan' calculator. The app has a purple header bar with a status bar at the top showing the time as 9:29 and various icons. The main content area is white with purple accents. It features five input fields for loan details: 'Enter Amount:' (5000), 'Interest Rate:' (4), 'Number of Repayments:' (12), and 'Started Date:' (2024-8-1). Below these fields are three purple buttons labeled 'Calculate', 'Back', and 'Convert'. At the bottom, there are three white boxes with purple borders displaying the calculated results: 'Monthly Installment: \$433.33', 'Last Payment Date: 2025-08-01', and 'Total Amount: \$5200.00'.

Field	Value
Enter Amount:	5000
Interest Rate:	4
Number of Repayments:	12
Started Date:	2024-8-1
Monthly Installment:	\$433.33
Last Payment Date:	2025-08-01
Total Amount:	\$5200.00

Figure 4

For example, if the user enters total amount RM5000, interest rate 4%, number of repayments 12 and the started date 1st August 2024. After clicking the “Calculate” button, the result of monthly instalment, last payment date and total amount will be shown at the bottom of the third button. (**Figure 4**)

Table of Amortization Schedule table (personal loan)



The screenshot shows a mobile application interface with a purple header bar. The title 'Amortization Schedule' is centered below the header. A table with three columns is displayed: 'Payment number', 'Beginning Balance (RM)', and 'Monthly I'. The table contains 12 rows of data. Below the table, the loan details are listed: 'Personal loan', 'Loan amount (RM): 5000.00', 'Interest rate (% per annum): 4.00', and 'Number of Repayments (months): 12'. A purple 'Back' button is located at the bottom of the screen.

Payment number	Beginning Balance (RM)	Monthly I
1	5000.00	16.67
2	4583.33	16.67
3	4166.67	16.67
4	3750.00	16.67
5	3333.33	16.67
6	2916.67	16.67
7	2500.00	16.67
8	2083.33	16.67
9	1666.67	16.67
10	1250.00	16.67
11	833.33	16.67
12	416.67	16.67

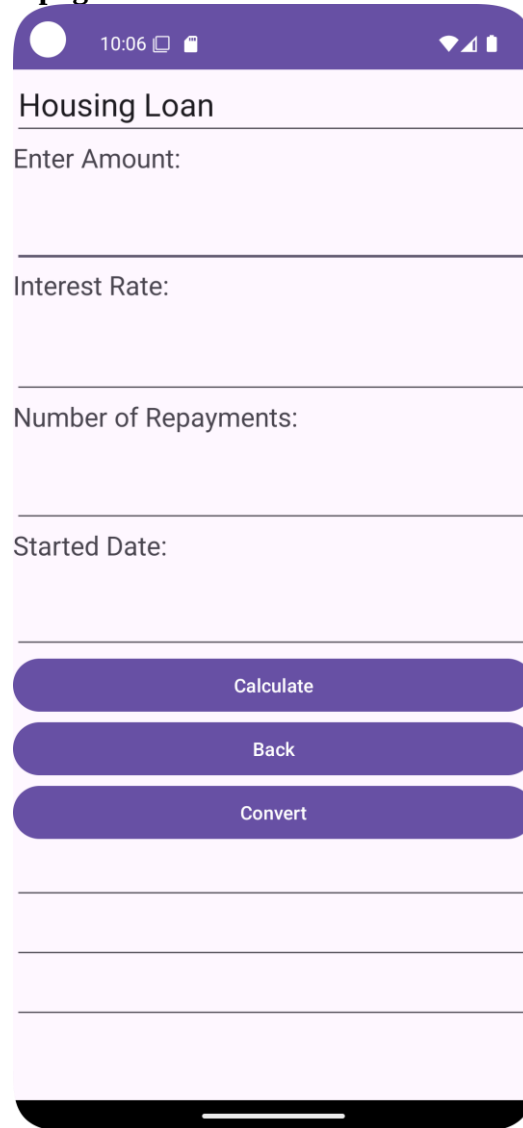
Personal loan
Loan amount (RM): 5000.00
Interest rate (% per annum): 4.00
Number of Repayments (months): 12

Back

Figure 5

When the user clicked on “Convert” button, it will lead them to the amortization schedule table shown above. This amortization schedule will show the payment number, beginning balance (RM), Monthly payment (RM), Interest paid (RM), and Principal paid (RM). This table have a function which is scrolling up and down, left and right. It has a “Back” button which can lead the user back to the personal loan in **Figure 4**.

Housing Loan Calculation page



Housing Loan

Enter Amount:

Interest Rate:

Number of Repayments:

Started Date:

Calculate

Back

Convert

Figure 6

When the user clicked on the Housing Loan button in **Figure 2**, it will proceed to housing loan calculation page. Then the user can enter the loan amount, interest rate, number of repayments and the started date which provides in **Figure 6**.

Housing Loan

Enter Amount:

5000

Interest Rate:

4

Number of Repayments:

12

Started Date:

2024-8-1

Calculate

Back

Convert

Monthly Installment: \$425.75

Last Payment Date: 2025-08-01

Total Amount: \$5108.99

Figure 7

For example, if the user enters total amount RM5000, interest rate 4%, number of repayments 12 and the started date 1st August 2024. After clicking the “Calculate” button, the result of monthly instalment, last payment date and total amount will be shown at the bottom of the third button. (**Figure 7**)

Table of Amortization Schedule table (housing loan)

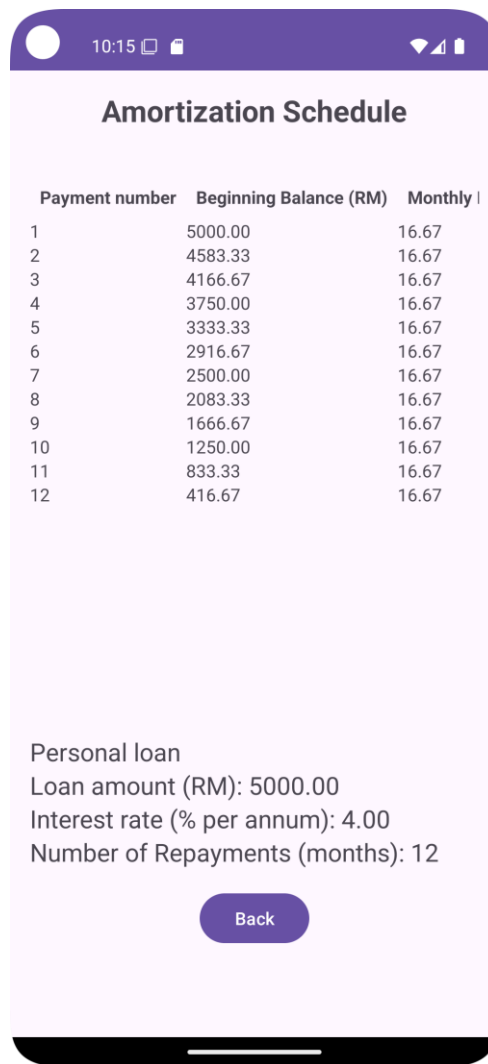


Figure 8

When the user clicked on “Convert” button, it will lead them to the amortization schedule table shown above. This amortization schedule will show the payment number, beginning balance (RM), Monthly payment (RM), Interest paid (RM), and Principal paid (RM). This table have a function which is scrolling up and down, left and right. It has a “Back” button which can lead the user back to the personal loan in **Figure 8**.

Appendix

Welcome to Loan Calculate.kt

```
1 package com.example.loancalculator
2
3 import android.content.Intent
4 import androidx.appcompat.app.AppCompatActivity
5 import android.os.Bundle
6 import android.widget.Button
7 import android.widget.EditText
8 import android.widget.Toast
9 import com.example.loancalculator.R.id.bt_nextpg
10 import com.example.loancalculator.R.id.tx_date
11
12 </> class MainActivity : AppCompatActivity() {
13     override fun onCreate(savedInstanceState: Bundle?) {
14         super.onCreate(savedInstanceState)
15         setContentView(R.layout.activity_main)
16
17         val dateBtn : Button! = findViewById<Button>(R.id.bt_nextpg)
18         dateBtn.setOnClickListener { // View!
19             val dateText : String = findViewById<EditText>(R.id.tx_date).text.toString()
20             val age : Int? = dateText.toIntOrNull()
21
22             if (dateText.isEmpty()) {
23                 Toast.makeText(context, this, text: "Please fill in your age!", Toast.LENGTH_SHORT).show()
24             } else if (age == null || age < 22 || age > 60) {
25                 Toast.makeText(context, this, text: "Please enter an age between 22 to 60", Toast.LENGTH_SHORT).show()
26             } else {
27                 Toast.makeText(context, this, text: "Submitted successfully!", Toast.LENGTH_SHORT).show()
28                 val intent1 = Intent(context, MainActivity2::class.java)
29                 startActivity(intent1)
30             }
31         }
32     }
33 }
34
```

Selection.kt

```
1 package com.example.loancalculator
2
3 import android.content.Intent
4 import androidx.appcompat.app.AppCompatActivity
5 import android.os.Bundle
6 import android.widget.Button
7
8 </> class MainActivity2 : AppCompatActivity() {
9     override fun onCreate(savedInstanceState: Bundle?) {
10         super.onCreate(savedInstanceState)
11         setContentView(R.layout.activity_main2)
12
13         var button2 : Button! = findViewById<Button>(R.id.tx_personalLoan)
14         button2.setOnClickListener() { // View!
15             val Intent2 = Intent(context, MainActivity3::class.java)
16             startActivity(Intent2)
17         }
18
19         var button3 : Button! = findViewById<Button>(R.id.tx_housingLoan)
20         button3.setOnClickListener() { // View!
21             val Intent3 = Intent(context, MainActivity4::class.java)
22             startActivity(Intent3)
23         }
24     }
25 }
26
```

Personal Loan.kt

```
1 package com.example.loancalculator
2 import android.content.Intent
3 import androidx.appcompat.app.AppCompatActivity
4 import android.os.Bundle
5 import android.widget.Button
6 import android.widget.EditText
7 import android.widget.TextView
8 import android.widget.Toast
9 import java.text.SimpleDateFormat
10 import java.util.Calendar
11 import java.util.Locale
12
13 <?> class MainActivity3 : AppCompatActivity() {
14     override fun onCreate(savedInstanceState: Bundle?) {
15         super.onCreate(savedInstanceState)
16         setContentView(R.layout.activity_main3)
17
18         val button4 : Button = findViewById(R.id.bt_back)
19         button4.setOnClickListener { it: View! -> {
20             Toast.makeText(context, this, text: "Back button clicked", Toast.LENGTH_SHORT).show()
21             val intent3 = Intent(packageContext, this, MainActivity2::class.java)
22             startActivity(intent3)
23         }}
24
25         val amountEditText: EditText = findViewById(R.id.line2)
26         val interestRateEditText: EditText = findViewById(R.id.line4)
27         val numInstallmentsEditText: EditText = findViewById(R.id.line6)
28         val startDateEditText: EditText = findViewById(R.id.line8)
29         val calculateButton: Button = findViewById(R.id.bt_calculate)
30         val monthlyInstallmentTextView: TextView = findViewById(R.id.tx_outputanswer)
31         val lastPaymentDateTextView: TextView = findViewById(R.id.line10)
32         val totalAmountTextView: TextView = findViewById(R.id.line12)
33
34         calculateButton.setOnClickListener { it: View! -> {
35             Toast.makeText(context, this, text: "Calculate button clicked", Toast.LENGTH_SHORT).show()
36             val amount : Double? = amountEditText.text.toString().toDoubleOrNull()
37             val interestRate : Double? = interestRateEditText.text.toString().toDoubleOrNull()
38             val numInstallments : Int? = numInstallmentsEditText.text.toString().toIntOrNull()
39             val startDate : String = startDateEditText.text.toString()
40
41             if (amount != null && interestRate != null && numInstallments != null && startDate.isNotEmpty()) {
42                 val monthlyInstallment : Double = calculateMonthlyInstallment(amount, interestRate, numInstallments)
43                 val lastPaymentDate : String = calculateLastPaymentDate(startDate, numInstallments)
44                 val totalAmount : Double = calculateTotalAmount(amount, interestRate, numInstallments)
45
46                 monthlyInstallmentTextView.text = "Monthly Installment: $%.2f".format(monthlyInstallment)
47                 lastPaymentDateTextView.text = "Last Payment Date: $lastPaymentDate"
48                 totalAmountTextView.text = "Total Amount: $%.2f".format(totalAmount)
49
50                 // Pass data to MainActivity5
51                 val intent = Intent(packageContext, this, MainActivity5::class.java)
52                 intent.putExtra(name: "loanAmount", amount)
53                 intent.putExtra(name: "interestRate", interestRate)
54                 intent.putExtra(name: "months", numInstallments)
55                 startActivity(intent)
56             } else {
57                 Toast.makeText(context, this, text: "Please fill in all fields correctly", Toast.LENGTH_SHORT).show()
58             }
59         }}
60     }
61
62     private fun calculateMonthlyInstallment(amount: Double, interestRate: Double, numInstallments: Int): Double {
63         val monthlyInterestRate : Double = (interestRate / 100) / 12
64         return (amount * (1 + (monthlyInterestRate * numInstallments))) / numInstallments
65     }
66
67     private fun calculateLastPaymentDate(startDate: String, numInstallments: Int): String? {
68         val sdf = SimpleDateFormat(pattern: "yyyy-MM-dd", Locale.getDefault())
69         val date : Date? = sdf.parse(startDate)
70         return if (date != null) {
71             val calendar : Calendar = Calendar.getInstance()
72             calendar.time = date
73             calendar.add(Calendar.MONTH, numInstallments)
74             sdf.format(calendar.time)
75         } else {
76             null
77         }
78     }
79
80     private fun calculateTotalAmount(amount: Double, interestRate: Double, numInstallments: Int): Double {
81         val monthlyInstallment : Double = calculateMonthlyInstallment(amount, interestRate, numInstallments)
82         return monthlyInstallment * numInstallments
83     }
84 }
```

Housing Loan.kt

```
1 package com.example.loancalculator
2
3 import android.content.Intent
4 import androidx.appcompat.app.AppCompatActivity
5 import android.os.Bundle
6 import android.widget.Button
7 import android.widget.EditText
8 import android.widget.TextView
9 import android.widget.Toast
10 import java.text.SimpleDateFormat
11 import java.util.Calendar
12 import java.util.Locale
13 import kotlin.math.pow
14
15 </> class MainActivity4 : AppCompatActivity() {
16     override fun onCreate(savedInstanceState: Bundle?) {
17         super.onCreate(savedInstanceState)
18         setContentView(R.layout.activity_main4)
19
20         val button5 : Button = findViewById<Button>(R.id.bt_back2)
21         button5.setOnClickListener { // View!
22             val intent4 = Intent( packageContext: this, MainActivity2::class.java)
23             startActivity(intent4)
24         }
25
26         val button7 : Button = findViewById<Button>(R.id.bt_gnrate2)
27         button7.setOnClickListener { // View!
28             Toast.makeText( context: this, text: "Convert button clicked", Toast.LENGTH_SHORT).show()
29             val intent6 = Intent( packageContext: this, MainActivity6::class.java)
30             startActivity(intent6)
31         }
32
33         val amountEditText: EditText = findViewById(R.id.line22)
34         val interestRateEditText: EditText = findViewById(R.id.line44)
35         val numInstallmentsEditText: EditText = findViewById(R.id.line66)
36         val startDateEditText: EditText = findViewById(R.id.line88)
37         val calculateButton: Button = findViewById(R.id.Calculatebutton)
38         val monthlyInstallmentTextView: TextView = findViewById(R.id.tx_outputswer2)
39         val lastPaymentDateTextView: TextView = findViewById(R.id.line100)
40         val totalAmountTextView: TextView = findViewById(R.id.line122)
41
42         calculateButton.setOnClickListener { // View!
43             Toast.makeText( context: this, text: "Calculate button clicked", Toast.LENGTH_SHORT).show()
44             val amount : Double? = amountEditText.text.toString().toDoubleOrNull()
45
46             val interestRate : Double? = interestRateEditText.text.toString().toDoubleOrNull()
47             val numInstallments : Int? = numInstallmentsEditText.text.toString().toIntOrNull()
48             val startDate : String = startDateEditText.text.toString()
49
50             if (amount != null && interestRate != null && numInstallments != null && startDate.isNotEmpty()) {
51                 val monthlyInstallment : Double = calculateMonthlyInstallment(amount, interestRate, numInstallments)
52                 val lastPaymentDate : String? = calculateLastPaymentDate(startDate, numInstallments)
53                 val totalAmount : Double = calculateTotalAmount(amount, interestRate, numInstallments) // Calculate total amount
54
55                 monthlyInstallmentTextView.text = "Monthly Installment: $%.2f".format(monthlyInstallment)
56                 lastPaymentDateTextView.text = "Last Payment Date: $lastPaymentDate"
57                 totalAmountTextView.text = "Total Amount: $%.2f".format(totalAmount)
58
59                 val intent = Intent( packageContext: this, MainActivity6::class.java)
60                 intent.putExtra( name: "loanAmount", amount)
61                 intent.putExtra( name: "interestRate", interestRate)
62                 intent.putExtra( name: "months", numInstallments)
63                 startActivity(intent)
64             } else {
65                 Toast.makeText( context: this, text: "Please fill in all fields correctly", Toast.LENGTH_SHORT).show()
66             }
67         }
68     }
69
70     private fun calculateMonthlyInstallment(
71         amount: Double,
72         interestRate: Double,
73         numInstallments: Int
74     ): Double {
75         val monthlyInterestRate : Double = (interestRate / 100) / 12
76         return if (monthlyInterestRate == 0.0) {
77             amount / numInstallments // Handle zero interest rate
78         } else {
79             val factor : Double = (1 + monthlyInterestRate).pow(numInstallments)
80             (amount * monthlyInterestRate * factor) / (factor - 1)
81         }
82     }
83
84     private fun calculateLastPaymentDate(startDate: String, numInstallments: Int): String? {
85         val sdf = SimpleDateFormat( pattern: "yyyy-MM-dd", Locale.getDefault())
86         val date : Date? = sdf.parse(startDate)
87     }
```

```

87         return if (date != null) {
88             val calendar :Calendar = Calendar.getInstance()
89             calendar.time = date
90             calendar.add(Calendar.MONTH, numInstallments)
91             sdf.format(calendar.time)
92         } else {
93             null
94         }
95     }
96
97     private fun calculateTotalAmount(amount: Double, interestRate: Double, numInstallments: Int): Double {
98         val monthlyInstallment :Double = calculateMonthlyInstallment(amount, interestRate, numInstallments)
99         return monthlyInstallment * numInstallments
100     }
101 }

```

Amortization Schedule (personal).kt

```

1 package com.example.loancalculator
2
3 import androidx.appcompat.app.AppCompatActivity
4 import android.os.Bundle
5 import android.widget.Button
6 import android.widget.TableLayout
7 import android.widget.TableRow
8 import android.widget.TextView
9
10 class MainActivity5 : AppCompatActivity() {
11     override fun onCreate(savedInstanceState: Bundle?) {
12         super.onCreate(savedInstanceState)
13         setContentView(R.layout.activity_main5)
14
15         val tableLayout: TableLayout = findViewById(R.id.amortizationTable)
16         val backBtn: Button = findViewById(R.id.backToCalculatorBtn)
17
18         val loanAmountGenerate: TextView = findViewById(R.id.loanAmountGenerate)
19         val interestGenerate: TextView = findViewById(R.id.interestGenerate)
20         val loanTenureGenerate: TextView = findViewById(R.id.loanTenureGenerate)
21
22         // Receive data from the intent
23         val principal :Double = intent.getDoubleExtra( name: "LoanAmount", defaultValue: 0.0)
24         val interestRate :Double = intent.getDoubleExtra( name: "interestRate", defaultValue: 0.0)
25         val months :Int = intent.getIntExtra( name: "months", defaultValue: 0)
26
27         // Display data
28         loanAmountGenerate.text = String.format("Loan amount (RM): %.2f", principal)
29         interestGenerate.text = String.format("Interest rate (%% per annum): %.2f", interestRate)
30         loanTenureGenerate.text = String.format("Number of Repayments (months): %d", months)
31
32         val amortizationSchedule :List<AmortizationMonth> = generateAmortization(principal, interestRate, months)
33
34         for (month :AmortizationMonth in amortizationSchedule) {
35             val row = TableRow( context: this)
36             val monthView = TextView( context: this)
37             val beginningBalanceView = TextView( context: this)
38             val interestPaidView = TextView( context: this)
39             val principalPaidView = TextView( context: this)
40             val monthlyPaymentView = TextView( context: this)
41
42             monthView.text = month.month.toString()
43             beginningBalanceView.text = String.format("%.2f", month.beginningBalance)
44             interestPaidView.text = String.format("%.2f", month.interestPaid)

```

```

45     principalPaidView.text = String.format("%.2f", month.principalPaid)
46     monthlyPaymentView.text = String.format("%.2f", month.monthlyPayment)
47
48     row.addView(monthView)
49     row.addView(beginningBalanceView)
50     row.addView(interestPaidView)
51     row.addView(principalPaidView)
52     row.addView(monthlyPaymentView)
53     tableLayout.addView(row)
54 }
55
56 backBtn.setOnClickListener { it: View?
57     finish()
58 }
59 }
60
61 private fun generateAmortization(principal: Double, interestRate: Double, months: Int): List<AmortizationMonth> {
62     val amortizationSchedule :MutableList<AmortizationMonth> = mutableListOf<AmortizationMonth>()
63     val monthlyRate :Double = (interestRate / 100) / 12
64     val monthlyPayment :Double = (principal + (principal * (interestRate / 100))) / months
65     val interestPaid :Double = principal * monthlyRate
66     val principalPaid :Double = monthlyPayment - interestPaid
67
68     var remainingPrincipal :Double = principal
69
70     for (i: Int in 1..months) {
71         val beginningBalance :Double = remainingPrincipal
72         remainingPrincipal -= principalPaid
73
74         amortizationSchedule.add(AmortizationMonth(i, beginningBalance, interestPaid, principalPaid, monthlyPayment))
75     }
76
77     return amortizationSchedule
78 }
79
80 data class AmortizationMonth(
81     val month: Int,
82     val beginningBalance: Double,
83     val interestPaid: Double,
84     val principalPaid: Double,
85     val monthlyPayment: Double
86 )

```

Amortization Schedule (housing).kt

```
1 package com.example.loancalculator
2
3 import androidx.appcompat.app.AppCompatActivity
4 import android.os.Bundle
5 import android.widget.Button
6 import android.widget.TableLayout
7 import android.widget.TableRow
8 import android.widget.TextView
9
10 class MainActivity : AppCompatActivity() {
11     override fun onCreate(savedInstanceState: Bundle?) {
12         super.onCreate(savedInstanceState)
13         setContentView(R.layout.activity_main)
14
15         val tableLayout: TableLayout = findViewById(R.id.amortizationTable)
16         val backBtn: Button = findViewById(R.id.backToCalculatorBtn2)
17
18         val loanAmountGenerate: TextView = findViewById(R.id.loanAmountGenerate2)
19         val interestGenerate: TextView = findViewById(R.id.interestGenerate2)
20         val loanTenureGenerate: TextView = findViewById(R.id.loanTenureGenerate2)
21
22         // Receive data from the intent
23         val principal: Double = intent.getDoubleExtra("loanAmount", 0.0)
24         val interestRate: Double = intent.getDoubleExtra("interestRate", 0.0)
25         val months: Int = intent.getIntExtra("months", 0)
26
27         // Display data
28         loanAmountGenerate.text = String.format("Loan amount (RM): %.2f", principal)
29         interestGenerate.text = String.format("Interest rate (% per annum): %.2f", interestRate)
30         loanTenureGenerate.text = String.format("Number of Repayments (months): %d", months)
31
32         val amortizationSchedule: List<AmortizationMonth> = generateAmortization(principal, interestRate, months)
33
34         for (month: AmortizationMonth in amortizationSchedule) {
35             val row = TableRow(context = this)
36             val monthView = TextView(context = this)
37             val beginningBalanceView = TextView(context = this)
38             val interestPaidView = TextView(context = this)
39             val principalPaidView = TextView(context = this)
40             val monthlyPaymentView = TextView(context = this)
41
42             monthView.text = month.month.toString()
43             beginningBalanceView.text = String.format("%.2f", month.beginningBalance)
44
45             interestPaidView.text = String.format("%.2f", month.interestPaid)
46             principalPaidView.text = String.format("%.2f", month.principalPaid)
47             monthlyPaymentView.text = String.format("%.2f", month.monthlyPayment)
48
49             row.addView(monthView)
50             row.addView(beginningBalanceView)
51             row.addView(interestPaidView)
52             row.addView(principalPaidView)
53             row.addView(monthlyPaymentView)
54             tableLayout.addView(row)
55         }
56
57         backBtn.setOnClickListener {
58             finish()
59         }
60     }
61
62     private fun generateAmortization(principal: Double, interestRate: Double, months: Int): List<AmortizationMonth> {
63         val amortizationSchedule: MutableList<AmortizationMonth> = mutableListOf<AmortizationMonth>()
64         val monthlyRate: Double = (interestRate / 100) / 12
65         val monthlyPayment: Double = (principal * (interestRate / 100)) / months
66         val interestPaid: Double = principal * monthlyRate
67         val principalPaid: Double = monthlyPayment - interestPaid
68
69         var remainingPrincipal: Double = principal
70
71         for (i: Int in 1..months) {
72             val beginningBalance: Double = remainingPrincipal
73             remainingPrincipal -= principalPaid
74
75             amortizationSchedule.add(AmortizationMonth(i, beginningBalance, interestPaid, principalPaid, monthlyPayment))
76         }
77
78         return amortizationSchedule
79     }
80
81     data class AmortizationMonth(
82         val month: Int,
83         val beginningBalance: Double,
84         val interestPaid: Double,
85         val principalPaid: Double,
86         val monthlyPayment: Double
87     )
88 }
```